A Protocol Stack and System for Parametric Data Networking

Reinaldo Javier Aguiar Marcano

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Title: A Protocol Stack and System for Parametric Data Networking

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Cross-Reference to Related Applications

This application is related to the inventions described in the U.S. Patent Applications for "A Central Processing Unit Architecture for Native Parametric and Symbolic Computation," "A Memory System for the Storage and Retrieval of Parametric and Symbolic Data Representations," and "A Method and System for High-Radix Physical Layer Data Encoding and Representation," the contents of which are hereby incorporated by reference in their entirety. The present invention utilizes the principles of said applications to create a complete networking paradigm.

Abstract

A network protocol stack and system for the transmission of data as parametric or symbolic representations rather than as conventional binary bit-streams. The protocol defines a packet structure wherein the payload consists of an unevaluated mathematical or symbolic "recipe" for data. Network nodes, such as routers and switches, are configured to route these parametric packets based on symbolic header information without needing to interpret or

evaluate the payload. This architecture allows data with effectively unlimited precision to traverse multiple network segments and be delivered to a destination node intact, with no loss of precision due to intermediate processing or network traversal. The final evaluation of the data occurs only at the destination application, preserving perfect fidelity across the network.

Background of the Invention

Conventional networking protocols, such as the TCP/IP suite, are fundamentally designed to transport streams of binary data. All information, regardless of its complexity or required precision, must first be serialized into a sequence of bits. This bit-stream is then segmented into packets, transmitted across a network, and reassembled at the destination. While robust, this model forces the premature evaluation and approximation of data. A high-precision number, for instance, must be converted to a standard floating-point representation before it can even be placed in a TCP packet, losing its perfect fidelity at the source. The network's sole function is to transport these already-approximated bits.

Summary of the Invention

The present invention provides a complete networking protocol stack and system designed to natively transport parametric and symbolic data. It redefines the concept of a data packet from a container of bits to a container for an unevaluated mathematical "recipe."

The system defines a new packet structure, the "parametric packet." This packet contains a header and a payload. The header may contain routing and control information in a symbolic or high-radix format. The payload is a variable-length field designed to hold a data structure representing a mathematical expression, such as '7/16' or a formula for calculating a value.

Network infrastructure, such as routers and switches, are equipped with a parametric forwarding engine. This engine reads the symbolic header of an incoming parametric packet and determines the appropriate outbound port based on its routing table, which is also indexed symbolically. Critically, the forwarding engine does not parse, evaluate, or modify the packet's payload. The unevaluated "recipe" passes through the node untouched.

This method ensures end-to-end precision. Data generated by a parametric CPU can be packetized, transmitted across a multi-hop network, and received by another parametric

system without any loss of fidelity. The evaluation to a numerical value, if needed, occurs only after the data has been successfully delivered to the target application.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram comparing a traditional network data flow with the parametric network data flow of the present invention.

FIG. 2 is a conceptual diagram of a parametric packet, illustrating its symbolic header and unevaluated payload.

Detailed Description of the Invention

Referring to **FIG. 1**, the fundamental difference in data flow is illustrated. In a traditional network (100), a high-precision number (110) is first approximated into a fixed-width binary format (115). This lossy data is placed in a standard IP packet, traverses a router (120), and arrives at the destination with its precision already compromised. In the parametric network (150), the same high-precision number is encapsulated as a symbolic "recipe" (160) into a parametric packet. The packet traverses a parametric router (170) which forwards it based on its header without inspecting the payload. The packet arrives at the destination with its perfect, unevaluated "recipe" (160) fully intact.

FIG. 2 shows the structure of a parametric packet (200). The packet consists of a header (210) and a payload (220). The header contains fields such as 'Symbolic Destination Address' (212) and 'Symbolic Source Address' (214). These fields contain high-radix or symbolic representations, not traditional IP addresses. The payload (220) is a variable-length container

holding the unevaluated data structure (230), for example, the "recipe" for a number. The packet transports the concept of the data, not its evaluated value.

Claims

What is claimed is:

1. A method for networking, comprising:
    - encapsulating a data structure representing a mathematical expression into the payload of a data packet;
    - prepending a header to the payload to form a parametric packet, wherein the header contains a destination address in a symbolic or high-radix format;
    - transmitting the parametric packet from a source node onto a network;
    - receiving the packet at an intermediate network node;
    - at the intermediate node, reading only the header to determine a forwarding path; and
    - forwarding the parametric packet, with its payload unmodified, towards the destination address.
2. The method of claim 1, wherein the mathematical expression is unevaluated during its traversal of the intermediate network node.
3. The method of claim 1, wherein the method is implemented across a protocol stack that mirrors the layers of the OSI model but is adapted for parametric data.
4. The method of claim 1, further comprising receiving the packet at a destination node and extracting the complete, unmodified data structure from the payload.
5. A network node, comprising:
    - one or more network interfaces for receiving and transmitting packets;
    - a routing table containing symbolic or high-radix addresses;
    - a parametric forwarding engine configured to:
        - receive a parametric packet containing a symbolic header and a payload;
        - extract a destination address from the symbolic header;

- query the routing table using the destination address to determine an outbound interface; and
- transmit the parametric packet via the outbound interface without evaluating or altering the payload.

6. The network node of claim 5, wherein the payload of the parametric packet contains a data structure representing an unevaluated mathematical expression.

7. A parametric data packet for transmission on a network, comprising:

   - a header field containing a destination address represented as a symbolic or high-radix value; and
   - a variable-length payload field configured to contain a data structure representing a complete, unevaluated mathematical expression.

Drawings

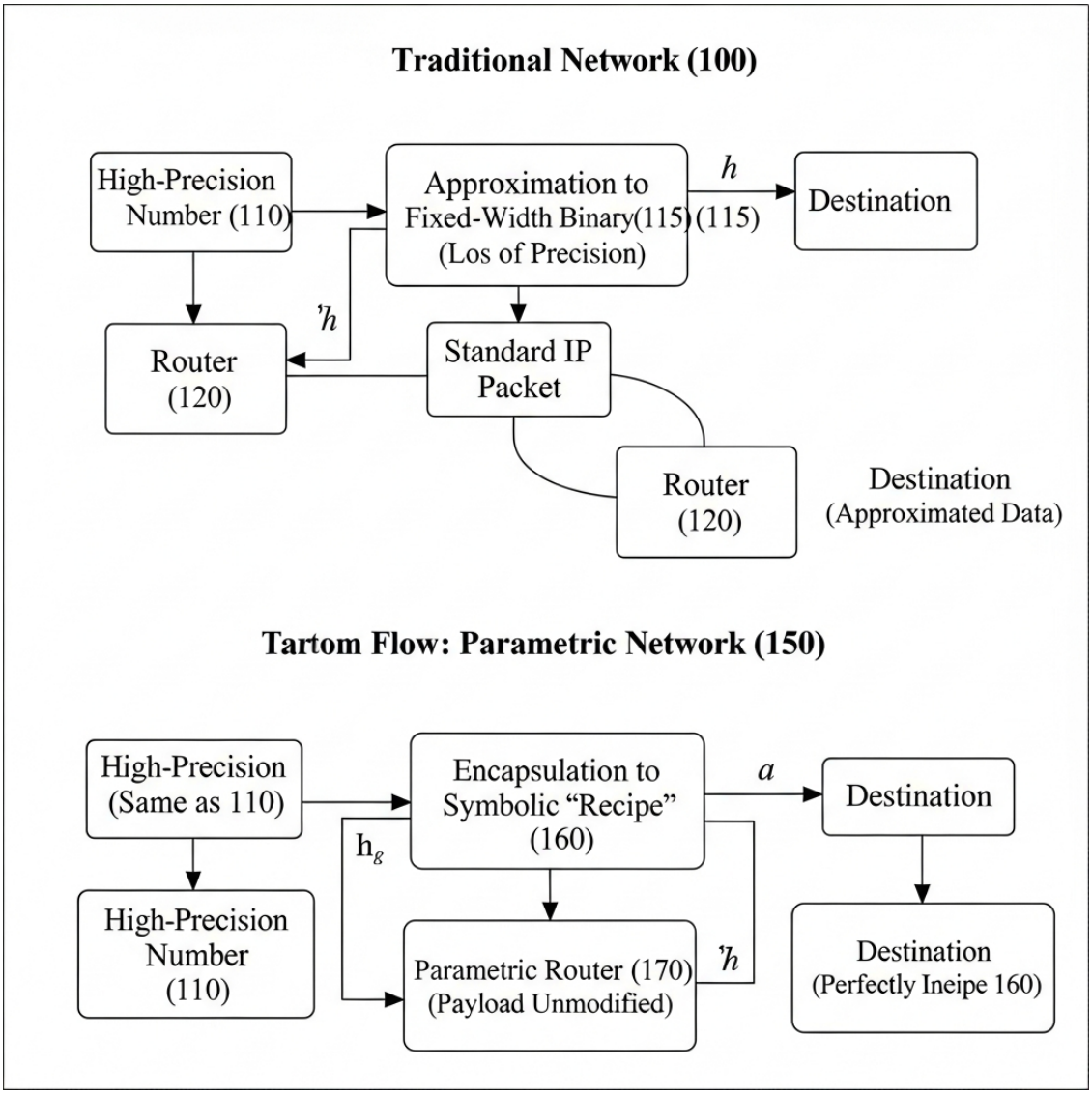


Figure 1: Block diagram comparing a traditional network data flow with the parametric network data flow of the present invention.

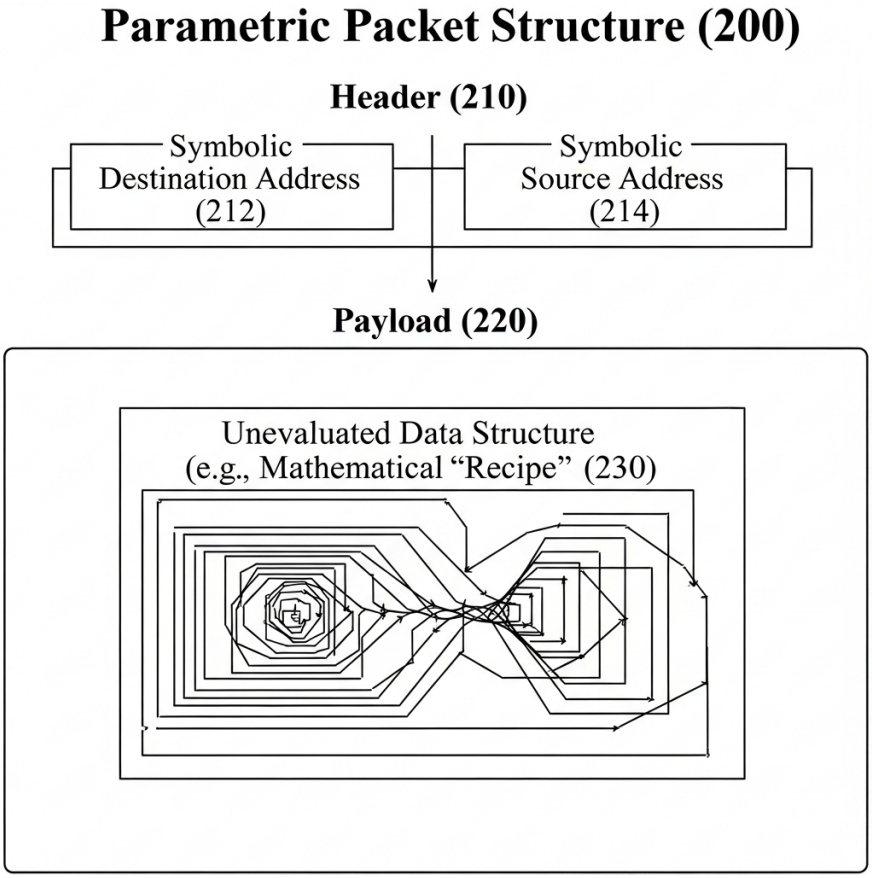


Figure 2: Conceptual diagram of a parametric packet, illustrating its symbolic header and unevaluated payload.